

1) Inclusion / Exclusion criteria

Inclusion criteria	Exclusion criteria
Must deal with community-acquired pneumonia	Not addressing community-acquired pneumonia
Must address HIV-infected population	Not addressing HIV-infected population.
Must include children aged 0-15 years	Includes children aged older than 15 years
Prospective, research studies, antemortem or postmortem studies, randomized controlled trials	Retrospective studies, systematic reviews, meta-analysis, clinical practice guidelines
Must include at least 20 cases	Includes less than 20 cases
Must assess Sub-Saharan countries	Not addressing Sub-Saharan African countries.
Must be published between 2009-2023	Published before 2009 or after 2023
Must be published in English, French, German, Portuguese, or Spanish	Published in other languages than English, French, German, Portuguese, Spanish
	Does not address the etiology of pneumonia

2) Search strategy

Two reviewers searched for indexed articles published in PubMed, Embase, Scopus, CENTRAL and Google Scholar databases that were published

between 01/02/2009 and 31/03/2023. The search was performed in English, French, German, Portuguese and Spanish and the option 'all fields' was used to recover the articles in which the search terms appeared in the title, abstract and/or keywords. Moreover, specific Emtree and Mesh terms for Embase and PubMed respectively were used.

Other strategies included the cross-checking of the reference list of all included articles for relevant studies and the review of summaries of relevant journals for studies about the topic.

Boolean query

The following search terms were screened in title, abstract and/or keywords using the AND Boolean logic operator:

(Pneumonia OR Pneumonias OR Lobar Pneumonia OR Lobar Pneumonias, Lobar OR Pneumonia, Lobar OR Pneumonitis OR Pneumonitides OR Pulmonary Inflammation OR Inflammation, Pulmonary OR Inflammations, Pulmonary OR Pulmonary Inflammations OR Lung Inflammation OR Inflammation, Lung OR Inflammations, Lung OR Lung Inflammations) AND

(HIV OR Human Immunodeficiency Virus OR Immunodeficiency Virus, Human OR Immunodeficiency Viruses, Human OR Virus, Human Immunodeficiency OR Viruses, Human Immunodeficiency OR Human Immunodeficiency Viruses OR AIDS OR AIDS Virus OR AIDS Viruses OR Virus, AIDS OR Viruses, AIDS OR Acquired Immune Deficiency Syndrome Virus OR Acquired Immunodeficiency Syndrome Virus) AND (Child OR Children OR Minors OR Child, Only OR Only Children OR Children, Only) AND

Africa South of the Sahara OR Angola OR Benin OR Botswana OR Burkina Faso OR Burundi OR Cabo Verde OR Cameroon OR Central African Republic OR Chad OR Comoros OR Congo OR Democratic Republic of Congo OR Cote d'Ivoire OR Djibouti OR Equatorial Guinea OR Eritrea OR Eswatini OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR Guinea-Bissau OR Kenya OR Lesotho OR Liberia OR Madagascar OR Malawi OR Mali OR Mauritania OR Mayotte OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR Sahel OR Sao Tome and Principe OR Senegal OR

Sierra Leone OR Somalia OR South Africa OR South Sudan OR
Sudan OR Tanzania OR Togo OR Uganda OR Zambia OR Zimbabwe

Databases

PubMed, Embase, Scopus, CENTRAL and Google Scholar databases

PubMed

("Pneumonia"[Mesh] OR pneumonia[tiab] OR pneumonias[tiab] OR pneumonitis[tiab] OR "Pulmonary Inflammation"[tiab] OR "Lung Inflammation"[tiab]) AND ("HIV"[Mesh] OR "HIV Infections"[Mesh] OR HIV[tiab] OR "Human Immunodeficiency Virus"[tiab] OR AIDS[tiab] OR "Acquired Immunodeficiency Syndrome"[tiab]) AND ("Child"[Mesh] OR "Infant"[Mesh] OR "Adolescent"[Mesh] OR child[tiab] OR children[tiab] OR infant*[tiab] OR pediatric*[tiab]) AND ("Africa South of the Sahara"[Mesh] OR "Sub-Saharan Africa"[tiab] OR Angola[tiab] OR Benin[tiab] OR Botswana[tiab] OR "Burkina Faso"[tiab] OR Burundi[tiab] OR "Cabo Verde"[tiab] OR Cameroon[tiab] OR "Central African Republic"[tiab] OR Chad[tiab] OR Comoros[tiab] OR Congo[tiab] OR "Democratic Republic of Congo"[tiab] OR "Cote d'Ivoire"[tiab] OR Djibouti[tiab] OR "Equatorial Guinea"[tiab] OR Eritrea[tiab] OR Eswatini[tiab] OR Ethiopia[tiab] OR Gabon[tiab] OR Gambia[tiab] OR Ghana[tiab] OR Guinea[tiab] OR "Guinea-Bissau"[tiab] OR Kenya[tiab] OR Lesotho[tiab] OR Liberia[tiab] OR Madagascar[tiab] OR Malawi[tiab] OR Mali[tiab] OR Mauritania[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR Niger[tiab] OR Nigeria[tiab] OR Rwanda[tiab] OR Senegal[tiab] OR "Sierra Leone"[tiab] OR Somalia[tiab] OR "South Africa"[tiab] OR "South Sudan"[tiab] OR Sudan[tiab] OR Tanzania[tiab] OR Togo[tiab] OR Uganda[tiab] OR Zambia[tiab] OR Zimbabwe[tiab]) AND (("2009/02/01"[Date - Publication] : "2023/03/31"[Date - Publication]))
("Pneumonia"[Mesh] OR pneumonia[tiab] OR pneumonias[tiab] OR pneumonitis[tiab] OR "Pulmonary Inflammation"[tiab] OR "Lung Inflammation"[tiab]) AND ("HIV"[Mesh] OR "HIV Infections"[Mesh] OR HIV[tiab] OR "Human Immunodeficiency Virus"[tiab] OR AIDS[tiab] OR "Acquired Immunodeficiency Syndrome"[tiab]) AND ("Child"[Mesh] OR "Infant"[Mesh] OR "Adolescent"[Mesh] OR child[tiab] OR children[tiab] OR infant*[tiab] OR pediatric*[tiab]) AND ("Africa South of the Sahara"[Mesh] OR "Sub-Saharan Africa"[tiab] OR Angola[tiab] OR Benin[tiab] OR Botswana[tiab] OR "Burkina Faso"[tiab] OR Burundi[tiab] OR "Cabo Verde"[tiab] OR Cameroon[tiab] OR "Central African Republic"[tiab] OR Chad[tiab] OR Comoros[tiab] OR Congo[tiab] OR "Democratic Republic of Congo"[tiab] OR "Cote d'Ivoire"[tiab] OR Djibouti[tiab] OR "Equatorial Guinea"[tiab] OR Eritrea[tiab] OR Eswatini[tiab] OR Ethiopia[tiab] OR Gabon[tiab] OR Gambia[tiab] OR Ghana[tiab] OR Guinea[tiab] OR "Guinea-Bissau"[tiab] OR Kenya[tiab] OR Lesotho[tiab] OR Liberia[tiab] OR Madagascar[tiab] OR Malawi[tiab] OR Mali[tiab] OR Mauritania[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR Niger[tiab] OR Nigeria[tiab] OR Rwanda[tiab] OR Senegal[tiab] OR "Sierra Leone"[tiab] OR Somalia[tiab] OR "South Africa"[tiab] OR "South Sudan"[tiab] OR Sudan[tiab] OR Tanzania[tiab] OR Togo[tiab] OR Uganda[tiab] OR Zambia[tiab] OR Zimbabwe[tiab]) AND (("2009/02/01"[Date - Publication] : "2023/03/31"[Date - Publication]))
)

Embase

('pneumonia'/exp OR pneumonia:ti,ab OR pneumonias:ti,ab OR pneumonitis:ti,ab OR 'pulmonary inflammation':ti,ab OR 'lung inflammation':ti,ab) AND ('human

immunodeficiency virus'/exp OR 'hiv infection'/exp OR hiv:ti,ab OR 'human immunodeficiency virus':ti,ab OR aids:ti,ab OR 'acquired immune deficiency syndrome':ti,ab) AND ('child'/exp OR 'infant'/exp OR 'adolescent'/exp OR child:ti,ab OR children:ti,ab OR infant*:ti,ab OR pediatric*:ti,ab) AND ('sub-saharan africa'/exp OR 'sub-saharan africa':ti,ab OR angola:ti,ab OR benin:ti,ab OR botswana:ti,ab OR 'burkina faso':ti,ab OR burundi:ti,ab OR 'cabo verde':ti,ab OR cameroon:ti,ab OR 'central african republic':ti,ab OR chad:ti,ab OR comoros:ti,ab OR congo:ti,ab OR 'democratic republic of congo':ti,ab OR 'cote d'ivoire':ti,ab OR djibouti:ti,ab OR 'equatorial guinea':ti,ab OR eritrea:ti,ab OR eswatini:ti,ab OR ethiopia:ti,ab OR gabon:ti,ab OR gambia:ti,ab OR ghana:ti,ab OR guinea:ti,ab OR 'guinea bissau':ti,ab OR kenya:ti,ab OR lesotho:ti,ab OR liberia:ti,ab OR madagascar:ti,ab OR malawi:ti,ab OR mali:ti,ab OR mauritania:ti,ab OR mozambique:ti,ab OR namibia:ti,ab OR niger:ti,ab OR nigeria:ti,ab OR rwanda:ti,ab OR senegal:ti,ab OR 'sierra leone':ti,ab OR somalia:ti,ab OR 'south africa':ti,ab OR 'south sudan':ti,ab OR sudan:ti,ab OR tanzania:ti,ab OR togo:ti,ab OR uganda:ti,ab OR zambia:ti,ab OR zimbabwe:ti,ab) AND [2009-2023]/py

Scopus

(TITLE-ABS-KEY ((pneumonia OR pneumonias OR pneumonitis OR "pulmonary inflammation" OR "lung inflammation") AND (HIV OR "human immunodeficiency virus" OR AIDS OR "acquired immune deficiency syndrome") AND (child OR children OR infant* OR pediatric*) AND ("sub-saharan africa" OR angola OR benin OR botswana OR "burkina faso" OR burundi OR "cabo verde" OR cameroon OR "central african republic" OR chad OR comoros OR congo OR "democratic republic of congo" OR "cote d'ivoire" OR djibouti OR "equatorial guinea" OR eritrea OR eswatini OR ethiopia OR gabon OR gambia OR ghana OR guinea OR "guinea bissau" OR kenya OR lesotho OR liberia OR madagascar OR malawi OR mali OR mauritania OR mozambique OR namibia OR niger OR nigeria OR rwanda OR senegal OR "sierra leone" OR somalia OR "south africa" OR "south sudan" OR sudan OR tanzania OR togo OR uganda OR zambia OR zimbabwe)) AND PUBYEAR > 2008 AND PUBYEAR < 2024)

Cochrane

(pneumonia OR pneumonias OR pneumonitis OR "pulmonary inflammation" OR "lung inflammation") AND (HIV OR "human immunodeficiency virus" OR AIDS OR "acquired immune deficiency syndrome") AND (child OR children OR infant* OR pediatric*) AND ("sub-saharan africa" OR angola OR benin OR botswana OR "burkina faso" OR burundi OR "cabo verde" OR cameroon OR "central african republic" OR chad OR comoros OR congo OR "democratic republic of congo" OR "cote d'ivoire" OR djibouti OR "equatorial guinea" OR eritrea OR eswatini OR ethiopia OR gabon OR gambia OR ghana OR guinea OR "guinea bissau" OR kenya OR lesotho OR liberia OR madagascar OR malawi OR mali OR mauritania OR mozambique OR namibia OR niger OR nigeria OR rwanda OR senegal OR "sierra leone" OR somalia OR "south africa" OR "south sudan" OR sudan OR tanzania OR togo OR uganda OR zambia OR zimbabwe) with Publication Year from 2009 to 2023 in Title Abstract Keyword

Google Scholar

pneumonia HIV pediatric "sub-Saharan Africa" mortality etiology; Filters 2009-2013

Only partially screened